INT104 Lab Report

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***Index Terms*—Spearman Correlation Matrix, Shapiro-Wilk Test, Robust Standardization, Logarithmic Transformation, Box-Cox Transformation, Principal Component Analysis, K-Means, DBSCAN, Gaussian Mixture Model (GMM), Confusion Matrix, Logistic Regression, Boosting-Based Classifiers, Transformer Classifier**

I. INTRODUCTION

This report proceeds in the order of Data Observation and Standardization (Task I), Unsupervised Learning (Task II), and Supervised Learning (Task III). In the data observation section, we first evaluate features not involved in feature combination (e.g., *Gender*) using the Spearman Correlation Matrix. Subsequently, we explain why Z-Score standardization is not adopted via the Shapiro-Wilk Test, ultimately selecting Robust Standardization, Logarithmic Transformation, and Box-Cox Standardization. In the unsupervised learning section, we first introduce the rationale for feature selection and feature combination, followed by two clustering evaluation methods: the label-based Adjusted Rand Index (ARI) and the label-free Davies-Bouldin Index (DBI). Based on these metrics, we test the performance of K-Means, DBSCAN, and Gaussian Mixture Model (GMM) clustering algorithms. In the supervised learning section, we conduct accuracy tests on three categories of classifiers—Logistic Regression, Boosting-based models, and Transformer classifier models—by incorporating more combined features. We analyze the characteristics of each model using confusion matrices. The final conclusion interprets key findings from preceding sections, such as differences in feature selection and standardization approaches.

II. TASK I & II: DATA OBSERVATION & DATA STANDARDIZE

There are 466 rows in the data set with no null or missing values. The data set contains 8 features, which are *Programme*, *Gender*,*Grade*, *Q1*, *Q2*, *Q3*, *Q4*, *Q5*. Check the correlations among the eight features using a **Spearman Correlation Matrix**.

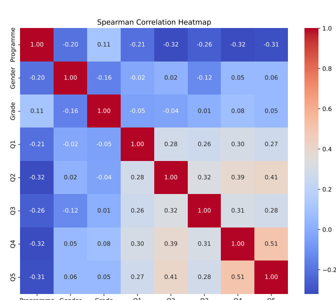

Fig. 1: Spearman correlation matrix

Obviously, the feature of *Programme* has almost **NO** correlation with other features. In addition, the features from *Q1* to *Q5* have a certain correlation, so it is deemed appropriate to conduct an integrated discussion of them as a unified entity. Additionally, when selecting features, consider **EXCLUDING** *Gender* because it is negatively correlated with *Programme*. Now 3 method will be utilized to standardize the data.

Considering that the distributions of *Q1*, *Q2*, *Q3*, *Q4*, and *Q5* may have outlier characteristics, **Robust** standardization is adopted. The median and interquartile range are used instead of the mean and standard deviation. It is more robust to outliers and can effectively reduce the impact of outliers on the standardization results. The formula is as follows.

$$x' = \frac{x - \text{median}(x)}{\text{IQR}(x)}, \text{IQR} = Q_3(x) - Q_1(x) \tag{1}$$

Where $Q_3(x)$ and $Q_1(x)$ represent the 75th percentile and the 25th percentile of the characteristic x, respectively. The result see **Fig. 2**. As a result,

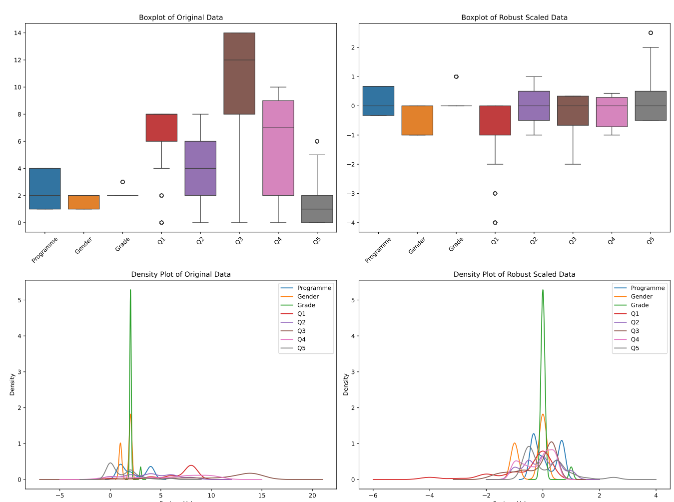

Fig. 2: Robust standardization.

Moreover, the **Shapiro-Wilk Test** is employed to check whether the original data (from *Q1* to *Q5*) conforms to the normal distribution, which determined us whether to apply **Z-score** standardization. The value of W indicates that closer to 1 means better normality and vise versa.

$$W = \frac{(\sum_{i-1}^{n} a_i x_{(i)})^2}{\sum_{i-1}^{n} (x_i - \overline{x})^2} \tag{2}$$

Assuming $\alpha = 0.05$, it was found that the p-values for *Q1* to *Q5* are all less than α.

TABLE I: Results of Shapiro-Wilk Normality Test for Features Q1 - Q5

Feature	Original Statistic	Original p	Robust Statistic	Robust p
Q1	0.6656	0.0000*	0.6656	0.0000*
Q2	0.9140	0.0000*	0.9140	0.0000*
Q3	0.7797	0.0000*	0.7797	0.0000*
Q4	0.8640	0.0000*	0.8640	0.0000*
Q5	0.7712	0.0000*	0.7712	0.0000*

* Indicates significant deviation from normality at the $\alpha = 0.05$ level.

Therefore, the null hypothesis is rejected, indicating that the data do **NOT** follow a normal distribution. Consequently, the Z-score standardization method is not applicable.

It can be seen from **Table I** that the data is severely skewed. Therefore, a logarithmic transformation can be performed first, and then normalization. See **Fig. 3**.

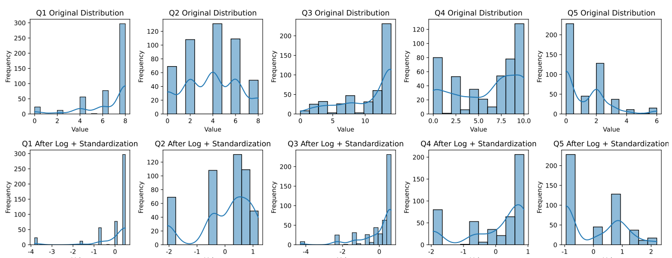

Fig. 3: Logarithmic standardization.

Similarly, using the Box-Cox transformation method followed by standardization. See **Fig. 4**.

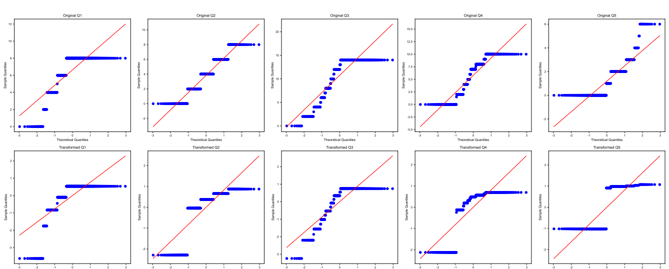

Fig. 4: Box-Cox standardization.

In general, for the data standardization process, the following 3 methods were used above:

- **Robust Standardization**, see **Fig. 2**.
- **Logarithmic Transformation + Normalization**, see **Fig. 3**.
- **Box-Cox Transformation + Normalization**, see **Fig. 4**.

Next, use the **Mean** and **Standard Deviation** as indicators of standardization. The evaluation method is as follows: calculate the mean and standard deviation for each feature of the data, and then take the average of the results. see the **Table II**.

Obviously, after standardization, the mean of the data is close to 0 and the standard deviation is close to 1. In conclusion, the standardization effect is better than that of the original data.

TABLE II: Evaluation of Standardize

Statistic	**Stanardize Method**			
	Original	**Robust**	**Logarithmic**	**Box-Cox**
Mean ($\bar{x}$)	4.280	-0.149	0.000	0.005
Standard Deviation (s)	1.988	0.601	1.000	1.000

III. Task III:Clustering

In this section, 2 evaluation methods are used for clustering, namely the **Davies Bouldin Index (DBI)** and the **Adjusted Rand Index (ARI)** [1], [2]. Among them, DBI is an evaluation method without true labels, while ARI is an evaluation method with true labels.

The **ARI** is mainly used to evaluate the similarity between the clustering result and the true classification. Its value ranges $[-1, 1]$. The higher the value, the more similar the clustering result is to the true classification.

$$ARI = \frac{\sum_{i,j}\binom{n_{ij}}{2} - \frac{\left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]}{\binom{n}{2}}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \frac{\left[\sum_i \binom{a_i}{2}\sum_j \binom{b_j}{2}\right]}{\binom{n}{2}}} \tag{3}$$

TABLE III: Notation Explanation of the ARI Formula

Notation	**Def.**
n	Total number of samples in the dataset
n_{ij}	Samples in i-th cluster of C and j-th cluster of K
a_i	Samples in i-th cluster of C
b_j	Samples in j-th cluster of K
$\binom{n}{k}$	Combination number, $\dfrac{n!}{k!(n-k)!}$

DBI evaluates the clustering result by calculating the ratio of the intra-cluster distance to the inter-cluster distance, and its formula is as follows.

$$DBI = \frac{1}{n}\sum_{i=1}^{n}\max_{j\neq i}\left(\frac{S_i + S_j}{d_{ij}}\right) \tag{4}$$

$$S_i = \frac{1}{|C_i|}\sum_{x\in C_i} d(x, c_i) \tag{5}$$

Next, perform a feature combination on the original data and the 3 standardization methods and use Principal Component Analysis (PCA) for dimensionality reduction. There are many combination methods, and one of them is listed here. For example,

- Feature I: Grade + Q1 + Q2

TABLE IV: Notation in the DBI Formula

Notation	Def.
n	No. of clusters
S_i	Int. compactness of i-th cluster
C_i	Sample set of i-th cluster
c_i	Center of i-th cluster
$d(x, c_i)$	Dist. from sample x to cluster center c_i
d_{ij}	$d_{ij} = d(c_i, c_j)$

- Feature II: Grade + Q3 + Q4
- Feature III: Grade + Q4 + Q5

The reason for not selecting *Programme* as one of the features is that when evaluating clustering using ARI, ground-truth labels are required—and *Programme* itself serves as those labels. Additionally, as shown in **Fig. 1**, *Programme* is nearly negatively correlated with all other features. Furthermore, the reason for choosing *Grade* over *Gender* as a feature is that *Gender* is negatively correlated with *Programme* whereas *Grade* shows the opposite (correlation with *Programme*). Directly merge these 3 sets of features and perform dimensionality reduction using PCA. The results are shown in **Fig. 5**.

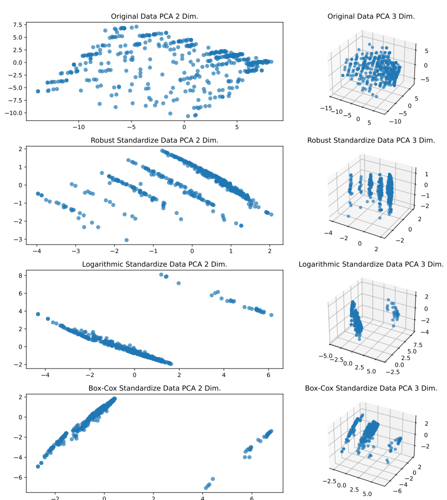

Fig. 5: Dimension reduction to 2 & 3 dim. using PCA

Next, 3 clustering methods will be used, namely **K-Means**, **DBSCAN** and **GMM** clustering.

The results of KMeans clustering are as follows: compared with the original data, the 3 standardized datasets significantly outperformed the original data. A value closer to 1 for ARI indicates higher consistency between the processed data and the ground-truth labels, while a value closer to 0 for DBI demonstrates better clustering performance in terms of compactness and separation.

Clearly, DBSCAN is less effective on raw, unprocessed data but performs better on data that has undergone 3 different transformations. The ARI values for the transformed data are higher than those for the original data, indicating a closer match with the actual classification. Furthermore, the DBI values for the transformed data are all lower than those for the original data, suggesting that the clustering models built from the transformed data are more optimal. See **Fig. 7** and **table VI**.

Next, we use Gaussian Mixture Model (GMM) clustering with n_components=4, and the results are as follows.

	ARI	DBI
Original Data KMeans val	0.087	0.391
Robust Standardize KMeans	0.099	0.337
Logarithmic Standardize KMeans val	0.191	0.200
Box-Cox Standardize KMeans val	0.181	0.193

TABLE V: K-Means ARI & DBI result

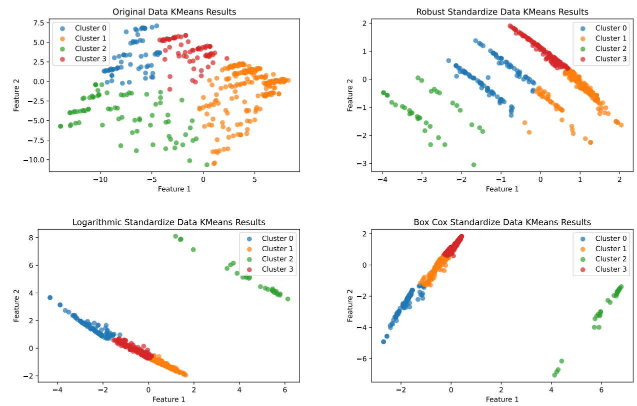

Fig. 6: K-Means cluster result

Data Type	EPS	Min_samples	ARI	DBI
Original Data	1.65	1	0.006	0.517
Robust Standardize	0.70	1	0.037	0.367
Logarithmic Standardize	0.80	1	0.106	0.190
Box-Cox Standardize	0.60	1	0.104	0.128

TABLE VI: Parameters and ARI & DBI result in DBSCAN

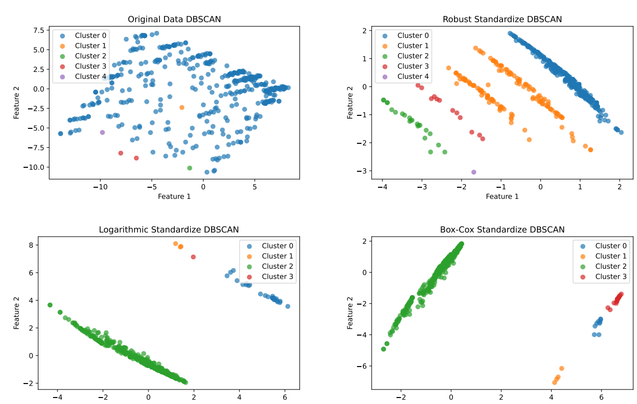

Fig. 7: DBSCAN cluster result

Data Type	ARI	DBI
Original Data	0.086	0.397
Robust Standardize	0.050	0.388
Logarithmic Standardize	0.206	0.151
Box-Cox Standardize	0.147	0.189

TABLE VII: GMM ARI & DBI result

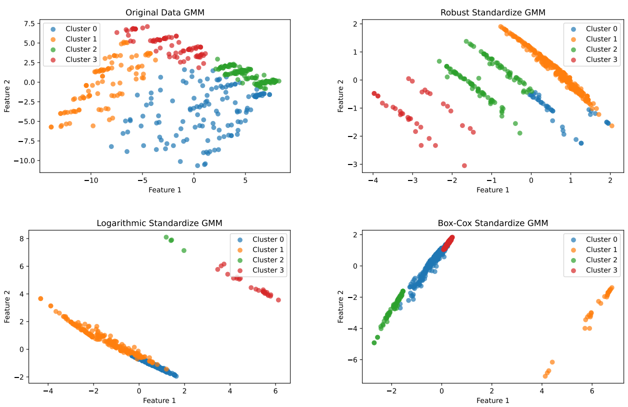

Fig. 8: GMM cluster result

IV. Task IV: Classification

In this section, we will attempt to use 3 different classifiers, which respectively belong to:

- Traditional machine learning classifiers—Logistic Regression
- Ensemble learning classifiers—Boosting-based models
- Deep learning classifiers—Artificial Neural Network (ANN)

Additionally, the training and test sets are split into an 7:3 ratio through random selection.

Compared to cluster analysis, classification prediction should incorporate more data to refine the model—specifically by using more sets of feature combinations. The data selection for the three models in this section is as follows:

TABLE VIII: Feature combination

Combination No.	**Features**
1	*Q1*, *Q2*, *Q3*, *Q4*, *Q5*
2	*Grade*, *Gender*
3	*Q1*, *Q3*
4	*Q2*, *Q4*
5	*Q3*, *Q5*
6	*Q1*, *Q2*, *Q3*
7	*Q2*, *Q3*, *Q4*
8	*Q3*, *Q4*, *Q5*
9	*Q4*, *Q5*
10	*Q1*, *Q5*
11	*Q1*, *Q2*

These combinations will also undergo 3 types of transformations (mentioned in Section II), specifically: **Robust Standardization**, **Logarithmic Standardization**, and **Box-Cox Standardization**. Subsequently, PCA will be used to reduce dimensionality to 1 dimension, which will then be merged into a matrix with 11 features to serve as input for the model. Additionally, the model is required to output four results representing the predicted *Programme*. The final result of accuracy is:

TABLE IX: Accuracy for Different Classifier under Different Transformation

Classifier \ Trans.	**Original**	**Robust**	**Logarithmic**	**Box-Cox**
Logistic Regression	52.85%	53.57%	**60.71%**	59.28%
Boosting-based	**65.00%**	58.57%	58.57%	62.85%
Transformer	57.85%	52.14%	58.57%	**64.28%**

Regarding the Logistic Regression Classifier (LR) model: the data processed through the three transformations demonstrated significantly higher accuracy. The dataset with the highest accuracy utilized Logarithmic Standardization. Therefore, it can be concluded that data preprocessing is necessary for traditional classification models (LR) and can, to a certain extent, improve the model's recognition rate. As shown in **Fig. 13**, the LR classifier frequently misclassifies *Programme* D as *Programme* A wrongly.

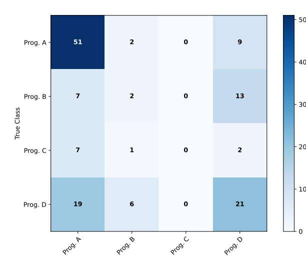

Fig. 9: Original

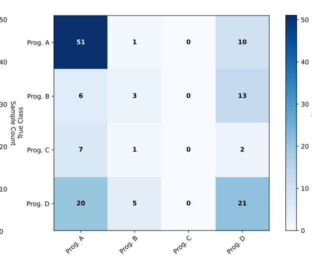

Fig. 10: Robust

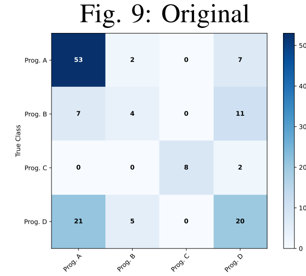

Fig. 11: Logarithmic

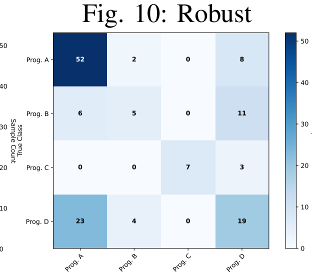

Fig. 12: Box-Cox

Fig. 13: Confusion Matrix of LR Classifier Model

For Boosting-based classifiers, the raw (unstandardized) data exhibits stronger performance. This may be because Boosting-based models are typically tree-based models, and a key characteristic of tree-based models is that they do not depend on feature scaling. Since they process split points of features rather than the absolute values of numerical values, they do not require standardization methods. In the confusion matrix, only the untreated data performed best. For the processed data, the number of correctly classified instances for *Programme* A was 47 in both cases, which may indicate that ensemble learning classifiers are insensitive to standardized data. See **Fig. 18** and **Table X**.

TABLE X: Parameters in Boosting-based Model

Parameters	**Value**
n_estimators	70
max_depth	1
learning_rate	0.4

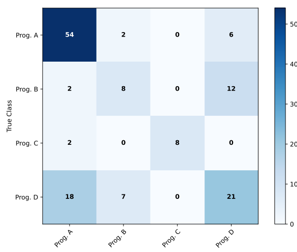

Fig. 14: Original

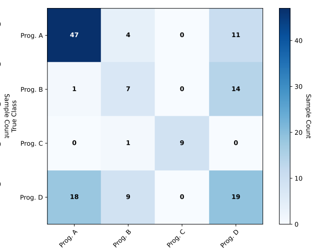

Fig. 15: Robust

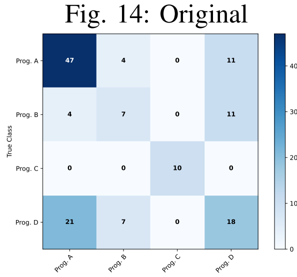

Fig. 16: Logarithmic

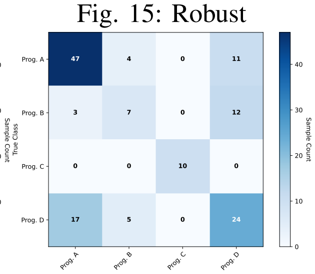

Fig. 17: Box-Cox

Fig. 18: Confusion Matrix of Boosting-based Classifier Model

TABLE XI: Parameters in Transformer Model

Parameters	**Value**
Input Dimension	11
Output Dimension	4
Hidden Layer	128
NHead	2
Num Layer	3
Epoch	150
Learning Rate	1e-3

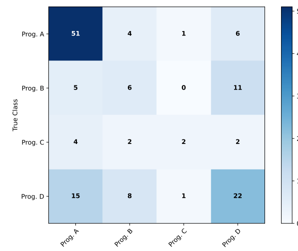

Fig. 19: Original

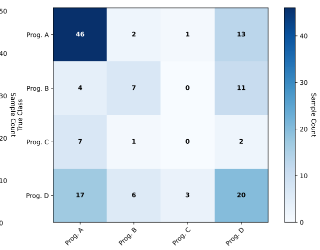

Fig. 20: Robust

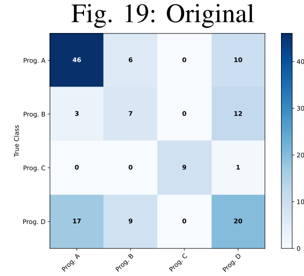

Fig. 21: Logarithmic

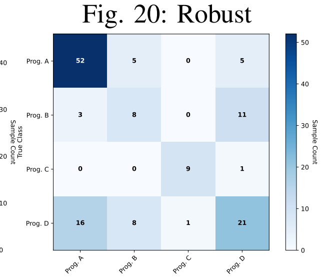

Fig. 22: Box-Cox

Fig. 23: Confusion Matrix of Transformer Classifier Model

Finally, in the Transformer classification model, data processed by Box-Cox Standardization exhibits exceptionally strong performance. To some extent, this is the opposite of Boosting-based models—i.e., deep learning classifiers perform better on standardized data than on non-standardized data. During training, the model was configured with an input layer of 11 units and an output layer of 4 units, where the optimal configuration for the hidden layers (after extensive tuning) was 128 units. For the batch size, 32 was deemed reasonable: too small a batch size (e.g., 8 or 4) would lead to overfitting and slower training, while a batch size of 64 would result in underfitting. A batch size of 32 strikes a balance between these extremes and exhibits strong data generalization capability. After repeated adjustments, the optimal learning rate was determined to be $1e-3$. With 150 epochs, the loss function stabilized around 0.69. It is worth noting that although the performance of the original data in the Transformer model is better than that of Robust, it can be clearly seen from the figure that its sensitivity (recognition rate) to *Programme* B is lower than the other three. Therefore, more factors need to be comprehensively considered. For the confusion matrix and model parameters, refer to **Fig. 23** and **TableXI**

V. Conclusion

There are some new discovery based on the unsupervised & supervised learning that mentioned before. In clustering, once the feature *Gender* is added to any feature combination, after many attempts, its clustering effect (based on the ARI index) is worse than that without adding this feature. This can be illustrated by **Fig. 1**: because the feature *Gender* is negatively correlated with the feature *Programme*. But it has no effect on the DBI index. The reason is that DBI is a label - free clustering evaluation, while ARI is a label - based clustering evaluation. The ratio of inter - cluster distance to intra - cluster distance in DBI will not be affected by negative correlation.

Secondly, it is found in classification (supervised learning) that traditional machine learning (logistic regression classification model) is sensitive to data standardization and often achieves better accuracy for those standardized data(see **Fig. IX.**). However, for Ensemble learning classifiers, they are not sensitive, because the accuracy of the original data is much higher than that of Robust and Logarithmic standardization, but only slightly higher than Box - Cox standardization. Furthermore, for the Transformer model, only the data standardized by Box - Cox perform well, while the data with other standardizations or no standardization perform poorly. Thus it is concluded that it is as sensitive to data standardization as traditional machine learning.

Furthermore, in both clustering and classification tasks, Robust standardization is generally found to be far less effective than

Logarithmic and Box-Cox standardization. A possible explanation is that when the data is skewed (as discussed in Section II)—for example, students have many high scores on certain Q items—logarithmic transformation adjusts the distribution to make the data more uniform. After standardization, this ensures more reasonable feature weights. Although Robust standardization is noise-resistant, it may over-compress the data and lose some discriminative information. However, it is precisely this unique information that allows the model to better identify which major a student belongs to. For instance, clustering algorithms like K-means make assumptions about data distribution, and performance improves when data is made more normally distributed. In classification models such as logistic regression, standardization accelerates gradient descent, while logarithmically transformed data may linearize relationships between features, enhancing model performance.

Finally, for future research, in the classification task, we can consider incorporating more models such as Random Forest and utilizing ensemble learning to enhance accuracy. In the clustering task, collecting more student profile data can be explored to improve the clustering performance measured by the ARI and DBI.

References

[1] L. Hubert and P. Arabie, "Comparing partitions," *Journal of classification*, vol. 2, pp. 193–218, 1985.

[2] D. L. Davies and D. W. Bouldin, "A cluster separation measure," *IEEE transactions on pattern analysis and machine intelligence*, no. 2, pp. 224–227, 2009.